CATHOLIC JUNIOR COLLEGE
General Certificate of Education Advanced Level
Higher 2
JC1 Promotional Examination

CANDIDATE NAME			
CLASS		INDEX NUMBER	

MATHEMATICS 9758/01

Paper 1

4 October 2021

3 hours

Candidates answer on the Question Paper.
Additional Materials: List of Formulae (MF26)

READ THESE INSTRUCTIONS FIRST

Write your name and class on all the work you hand in.

Write in dark blue or black pen on both sides of the paper.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** the questions.

Write your answers in the spaces provided in the Question Paper.

Give non-exact numerical answers correct to 3 significant figures, or 1 decimal place in the case of angles in degrees, unless a different level of accuracy is specified in the question.

You are expected to use an approved graphing calculator.

Unsupported answers from a graphing calculator are allowed unless a question specifically states otherwise.

Where unsupported answers from a graphing calculator are not allowed in a question, you are required to present the mathematical steps using mathematical notations and not calculator commands.

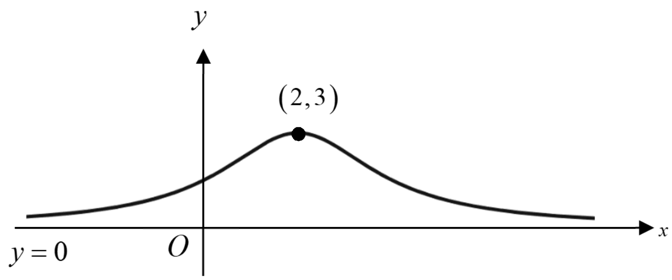
You are reminded of the need for clear presentation in your answers.

The number of marks is given in brackets [ ] at the end of each question or part question.

Question	**1**	**2**	**3**	**4**	**5**	**6**	**7**	**8**	**9**	**10**	**11**	**12**	**Total**
Marks													
Total	**3**	**4**	**7**	**8**	**9**	**8**	**9**	**7**	**11**	**12**	**11**	**11**	**100**

This document consists of **24** printed pages and **2** blank pages.

1.



The diagram shows the curve $y = \mathrm{f}(x)$. The curve has an asymptote $y = 0$ and a maximum point at $(2,3)$. It is given that f is concave downwards for $1 \le x \le 3$.

Sketch the graph of $y = \mathrm{f}'(x)$, stating the equations of any asymptotes, the $x-$coordinates of any stationary points and any points of intersection with the $x-$axis. [3]

2. In a convergent geometric series that only consists of positive terms, the sum of the first four terms is 80 times the sum of all the remaining terms. Find the common ratio. [4]

3. Differentiate the following expressions with respect to *x*, leaving your answers in terms of *x*,

(a) $\cos^{-1}\sqrt{x^3}$, giving your answer in non-trigonometric form, [2]

(b) $\operatorname{cosec}\left(\ln\left(x^2+3\right)\right)$, [2]

(c) $\left(x^2+1\right)^x$. [3]

4. The points $A(1,0,-2)$, $B(3,-1,-2)$ and $C(-3,7,0)$ lie on plane p_1. Another plane p_2 has equation $3x - y + 2z = 3$.

(i) Find a vector equation of plane p_1 in the form $\mathbf{r}\cdot\mathbf{n} = d$. [3]

(ii) Find the acute angle between p_1 and p_2. [2]

The equation of plane p_3 is given to be $-9x + 3y - 6z = 7$.

(iii) Find the shortest distance between p_2 and p_3. [3]

5. (a) (i) Express $\dfrac{2r^2+1}{r^2-1}$ in the form $A + \dfrac{B}{r-1} + \dfrac{C}{r+1}$, where *A*, *B* and *C* are constants to be found. [2]

(ii) Hence find $\displaystyle\sum_{r=2}^{n} \frac{2r^2+1}{r^2-1}$ in terms of *n*. [4]

(b) Express $\displaystyle\sum_{r=1}^{2n}\left[(-1)^{r+1}2^r\right]$ in the form $\dfrac{p}{q}\left[s\left(4^n\right)+t\right]$, where *p*, *q* ,*s* and *t* are integers to be determined. [3]

6. The curve C is defined by the parametric equations

$$x = \theta + \frac{1}{2}\sin 2\theta,\ y = 2\tan\theta \qquad \text{where } -\frac{\pi}{2} < \theta < \frac{\pi}{2}.$$

(i) Show that $\dfrac{dy}{dx} = \dfrac{1}{\cos^k \theta}$, where k is an integer to be determined. [3]

The lines T and N are the tangent and normal to C at the point where $\theta = \dfrac{\pi}{4}$ respectively.

(ii) Find the equations of T and N, leaving your answers in exact form. [3]

(iii) Find the area enclosed by T, N and the y-axis. [2]

7. It is given that $y = \sin\left[\ln(1+2x)\right]$.

(i) Show that $(1+2x)^2 \dfrac{d^2y}{dx^2} + 2(1+2x)\dfrac{dy}{dx} + ky = 0$, where k is a constant to be found.

Hence, find the first three non-zero terms of the Maclaurin expansion for y. [6]

(ii) Using standard series from the List of Formulae (MF26), verify the correctness of your result from part **(i)** up to and including the term in x^3. [2]

Explain why the expansion is not valid when $x = -\dfrac{1}{2}$. [1]

8 **(i)** The arithmetic progression is grouped into sets of integers, such that the n^{th} set contains n integers as shown.

$$\{\ 1\ \}, \{\ 3, 5\ \}, \{\ 7, 9, 11\ \}, \{\ 13, 15, 17, 19\ \}, \ldots \ldots$$

(a) Find the total number of terms in the first n sets, and hence show that the last term of the n^{th} set is $n^2 + n - 1$. [2]

(b) Find the first term of the n^{th} set. [2]

(c) Show that the sum of the terms in the n^{th} set is n^3. [1]

(ii) Hence, prove that $1^3 + 2^3 + 3^3 + \ldots\ldots + n^3 = \dfrac{1}{4}n^2(n+1)^2$. [2]

9. With reference to the origin O, the points A and B have position vectors $\mathbf{a} = -\mathbf{i} + \mathbf{j} + 2\mathbf{k}$ and $\mathbf{b} = 2\mathbf{j} + 5\mathbf{k}$ respectively.

(i) Find a vector equation of the line l_1 that passes through point A and is parallel to the vector **a**. [1]

(ii) Find the exact length of projection of **b** on l_1. Hence find d, the exact perpendicular distance from the point B to l_1. [4]

(iii) Using the value of d found in part **(ii)**, find the position vector of the point C, the foot of perpendicular from the point B to l_1. [3]

(iv) The line l_2 passes through point B and is parallel to vector **b**. Find a cartesian equation of l_3 which is the reflection of l_2 in l_1. [3]

10. The curves C_1 and C_2 have equations $y=\dfrac{2x^2+9}{x^2-4}$ and $\dfrac{x^2}{9}+\dfrac{y^2}{25}=1$ respectively.

(i) Find the equations of the asymptotes of the curve C_1. [3]

(ii) Sketch C_1 and C_2 on the same diagram, stating the equations of any asymptotes, coordinates of any points where C_1 or C_2 crosses the axes and any turning points. [5]

(iii) Find the x-coordinates of the points where the two curves intersect. [2]

(iv) Hence solve the inequality $-5\sqrt{1-\dfrac{1}{9}x^2}\le\dfrac{2x^2+9}{x^2-4}$. [2]

11. Sigmoid functions are used to model many natural processes such as population growth of virus. One example of a Sigmoid function f is given by

$$\mathrm{f}: x \mapsto \frac{1}{1+\mathrm{e}^{-x}}, \quad x\in\mathbb{R}.$$

(i) Sketch the graph of $y=\mathrm{f}(x)$, indicating clearly the equation(s) of any asymptote(s) and the coordinates of any points where the curve crosses the axes. [2]

(ii) Find $\mathrm{f}^{-1}(x)$ in similar form. [3]

Another function g is given by $\mathrm{g}: x\mapsto 3x-1,\ x\in\mathbb{R},\ 0\le x\le 2$.

(iii) Show that fg exists and find the range of fg, expressing your answer in terms of e. [4]

(iv) Describe a sequence of transformations which transform the graph of $y=\mathrm{f}(x)$ onto the graph of $y=\mathrm{fg}(x)$. [2]

12. (i) Amanda wants to make a mask holder. To do so, she cuts out 4 identical equilateral triangles of sides 1 cm and a rectangular strip $\dfrac{x-1}{16}$ cm by $\dfrac{2y-1}{4}$ cm from a rectangle of sides x cm by y cm (see Diagram A). The rectangular strip has an area of 5.5 cm^2.

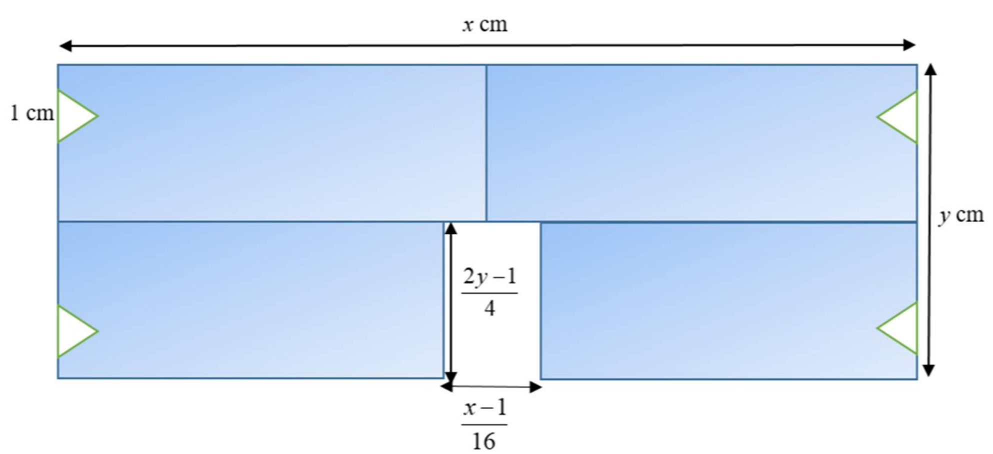


Diagram A

Show that A, the area of the mask holder is given by

$$A=\frac{176x}{x-1}+\frac{x}{2}-\sqrt{3}-\frac{11}{2}$$

Using differentiation, find the exact value of x such that area of the mask holder is a minimum. Hence determine the minimum area of the mask holder. [9]

(ii) The area of the rectangular strip is kept at 5.5 cm^2. Beth suggests that if the 4 identical triangle cut-outs are isosceles instead, the area of the mask holder could be made smaller as compared to the area found in part **(i)(b).**

For each of the triangles, let θ be the angle between two edges of length 1 cm each (see diagram B). Determine if Beth's suggestion is correct.

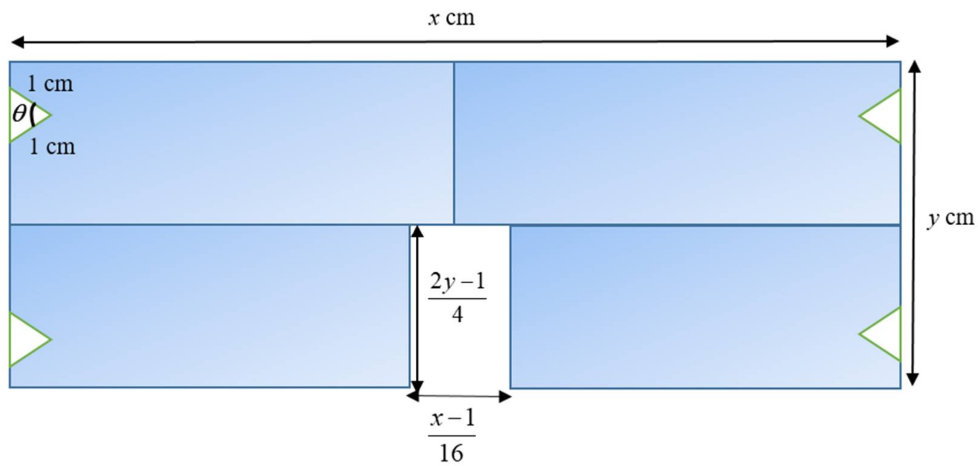


Diagram B [2]

End of Paper